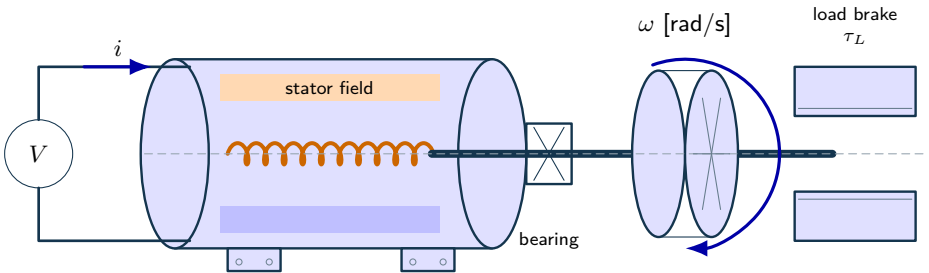

Electrical channel

$L\dot{i} = V - Ri - k_e\omega$.
Back emf opposes the applied voltage; input V is in volts.

Mechanical channel

$J\dot{\omega} = k_t i - b\omega - \tau_L$.
The rotation arrow is centered on the flywheel axis. Positive load torque opposes positive speed.

Energy conversion

$k_t i$ is torque [N m]; ω is speed [rad/s]. Bearing and brake details add no unmodeled dynamic states.